

FAST LOOKUP WITH HASHTABLES

Problem Solving with Computers-II

C++

```
#include <iostream>
using namespace std;

int main(){
    cout<<"Hola Facebook\n";
    return 0;
}
```

Hashtable for fast lookup

All operations $O(1)$ on average*:

- find
- insert
- erase

* no worst case guarantee but fast in practice!

// Just the essentials!

```
std::unordered_set<int> s = {1, 2, 3};  
s.insert(4);  
s.erase(2);  
bool found = s.count(3); // true
```



Source: <https://medium.com/@saurabh.bhoy910/stl-unordered-map-hashtable-c7054b28d07f>

Amazon User Tracking Scenario

Amazon engineers track ~200M unique users by IP addresses during peak shopping season. Each website access requires checking if the IP is in the list; if not, it's added. This helps analyze user interest and personalize recommendations.



Which data structure should Amazon engineers use to track ~200M unique users by IP addresses?

Goal - Fast search (lookup) and insert

Which data structure should Amazon engineers use to track ~200M unique users by IP addresses?

- Keys: IP addresses (32 bits)
 - example : 192.168.1.6
- $2^{32} \sim 4.3\text{B}$ possible IPs
- 200M (4.7%) unique users

192	168	1	6
(8 bits)	(8 bits)	(8 bits)	(8 bits)

192.168.1.6 → uint32_t: 3234251782

Can we achieve $O(1)$ search?

Naïve approach: 2^{32} -sized vector

Which data structure should Amazon engineers use to track ~200M unique users by IP addresses?

If you had to track unique IPs out of 4.3 billion (2^{32}) possibilities, how much memory do you think you'd need if you used a direct index approach?

2^{10} bytes = 1KB, 2^{20} bytes = 1 MB, 2^{30} bytes = 1GB

Assuming 1 bit per IP, we need 2^{32} bits = 2^{29} bytes = 512 MB

Assuming 4 bytes per IP = $512 * 32$ MB = 16 GB

Setup for hash tables

Universe of possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

- Keep track of evolving set S whose size is much less than the universe of all possible keys
- For example, 200M unique users (~ 5% of all possible IPs)

0	
1	
2	Key
3	
4	
5	
6	
7	
8	
9	

Insert key 192.168.1.6

- Keep track of evolving set S whose size is much less than the universe of all possible keys
- For example, 200M unique users (~ 5% of all possible IPs)

Universe of possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

192.168.1.6

Hash function: $h(x)$
Table size: m
Num keys: n

$h(x)$

$$\begin{aligned}h(x) &= \text{toint}(x) \bmod m \\&= \text{toint}(192.168.1.6) \bmod 10 \\&= 3234251782 \bmod 10 \\&= 2\end{aligned}$$

0	
1	
2	192.168.1.6
3	
4	
5	
6	
7	
8	
9	

Insert key 0.0.0.11

Universe of
possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

0.0.0.11

Hash function: $h(x)$
Table size: m
Num keys: n

$h(x)$

$$h(x) = \text{to_int}(x) \bmod m$$
$$= ?$$

0	
1	
2	192.168.1.6
3	
4	
5	
6	
7	
8	
9	

Insert key 0.0.0.11

Universe of possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

0.0.0.11

Hash function: $h(x)$
Table size: m
Num keys: n

$h(x)$

$$\begin{aligned}h(x) &= \text{toint}(x) \bmod m \\ &= 11 \bmod 10 \\ &= 1\end{aligned}$$

0	
1	0.0.0.11
2	192.168.1.6
3	
4	
5	
6	
7	
8	
9	

Insert 0.0.1.5

Universe of
possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

0.0.1.5

Hash function: $h(x)$
Table size: m
Num keys: n

$h(x)$

$$h(x) = \text{toint}(x) \bmod m \\ = ?$$

0	
1	0.0.0.11
2	192.168.1.6
3	
4	
5	
6	
7	
8	
9	

Insert key 0.0.1.5 results in a collision!

What should happen when two IPs hash to the same index?

A B C D
Crash Overwrite Chain Reject

Universe of
possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

0.0.1.5

$h(x)$

Hash function: $h(x)$
Table size: m
Num keys: n

$$\begin{aligned} h(x) &= \text{toint}(x) \bmod m \\ &= (256 + 5) \bmod 10 \\ &= 261 \bmod 10 \\ &= 1 \end{aligned}$$

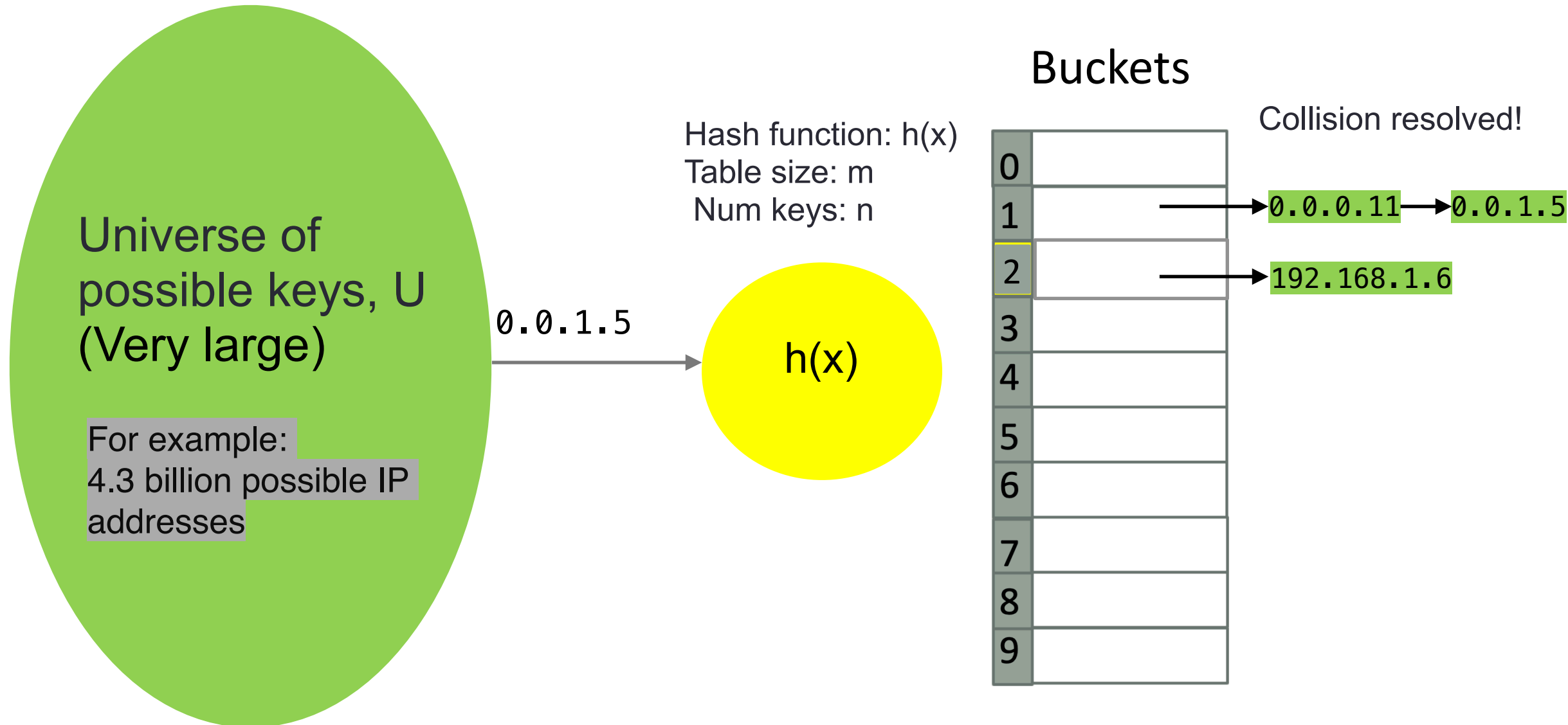
0	
1	0.0.0.11
2	192.168.1.6
3	
4	
5	
6	
7	
8	
9	



0.0.1.5

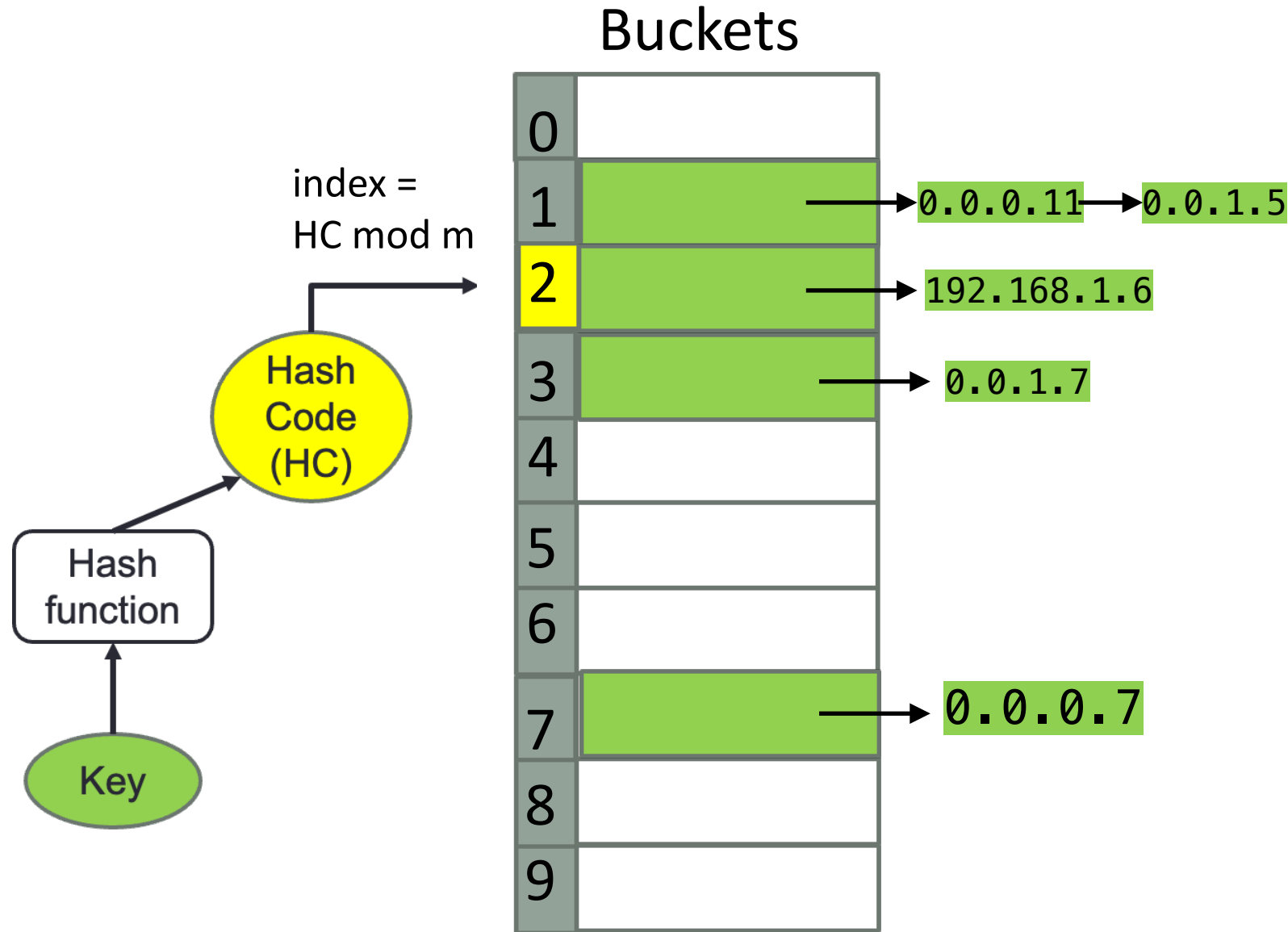
Collision: Two
keys map to the
same spot!

Resolving collisions: separate chaining



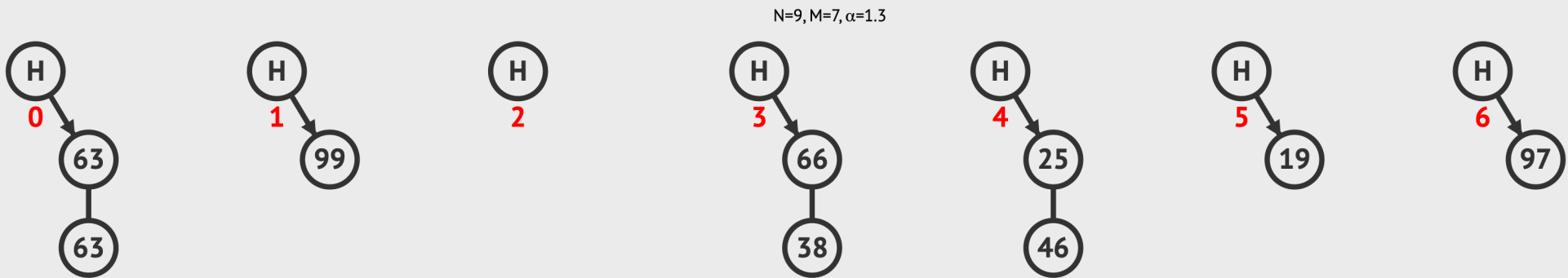
(Refined) Logical model of a hash table

- Keys stored in buckets (vector)
- Keys used to compute index of position in vector
- Each bucket can store multiple keys as a linked list
- Hashtable with separate chaining: Vector of linked list



Hashtable visualization

<https://visualgo.net/en/hashtable>



Create(M, N)

Search(v)

Insert(v)

Remove(v)

v =

8

Go

Goal - Fast search (lookup) and insert

Which data structure should Amazon engineers use to track ~200M unique users by IP addresses?

- A) Set (Balanced BST)
- B) unordered_set (Hashtable)
- C) Priority Queue
- D) Queue
- E) Vector with 2^{32} entries (one for each possible IP address)

192	168	1	6
(8 bits)	(8 bits)	(8 bits)	(8 bits)

192.168.1.6 → uint32_t: 3234251782

Total IPs: $2^{32} \approx 4.3\text{B}$

Design challenges

- **Deciding on collision resolution strategy**
- **Deciding the size of hash table**
- **Deciding the hash function**

Universe of
possible keys, U
(Very large)

For example:
4.3 billion possible IP
addresses

Keep track of evolving set S
whose size is much less than the
universe of all possible keys

0	
1	
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3	
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6	
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9	

For example, 200M unique users
(~ 5% of all possible IPs)

Can we guarantee good performance?

We implemented a hashtable with separate chaining.

Table size: m

Number of keys: n

Load factor $\alpha = n/m$

Hash function: $h(x) = x \bmod m$

If randomly uniformly selected keys, then “on average” things seem fine.

Expected time to search is $O(\alpha)$ (in practice, choose $\alpha = 1$)

Can all keys still hash to the same bucket?

(Worst-case scenario, linear search complexity!)

What's the chance of linear search complexity?

What is the probability that *all* keys hash to the *same* bucket?

Let's say we're inserting **$n = 1000$** keys into an empty table with **$m = 1000$** buckets.

A. 0

B. $1/1000$

C. $1/1000^{999}$

D. $1/1000^{1000}$

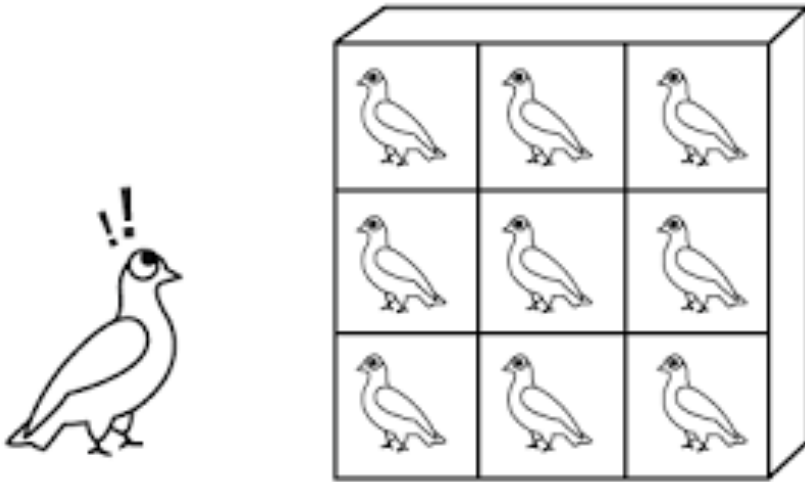
E. It depends...

- We're using the hash function **$h(x) = x \bmod m$**
- Keys are chosen independently and uniformly at random
- Hash table uses separate chaining

Is there a hash function that avoids linear search complexity?

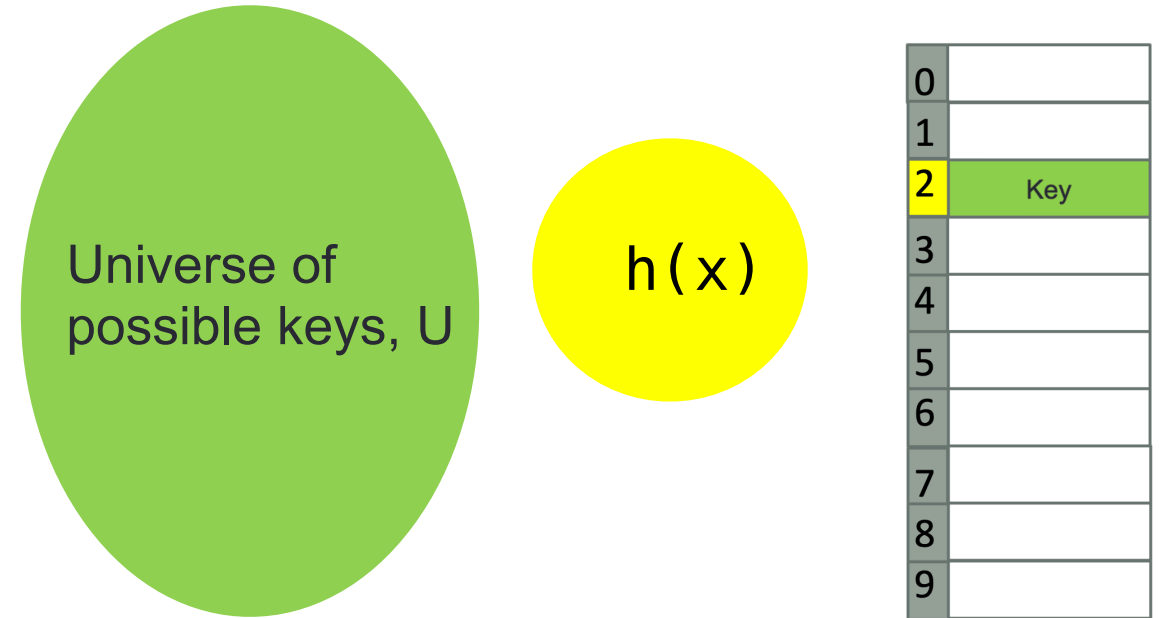
Is there a hash function that avoids linear search complexity?

THE PIGEONHOLE PRINCIPLE



Pigeonhole Principle:

If there are m pigeonholes and $m + 1$ pigeons, at least one pigeonhole has more than 1 pigeon



Generalized Pigeonhole Principle:

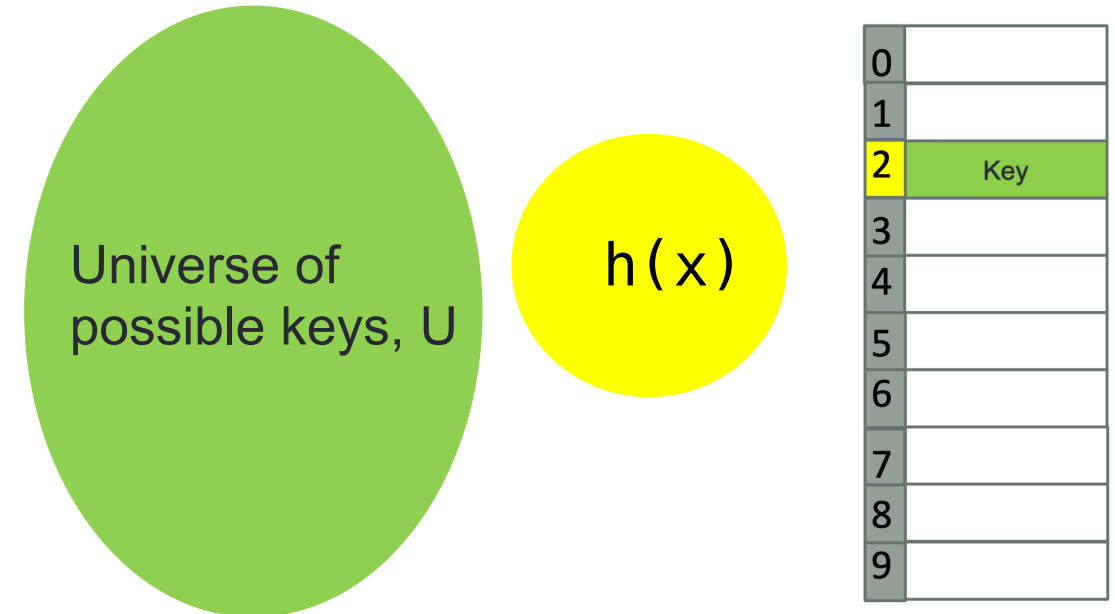
If there are m pigeonholes and $nm + 1$ pigeons, at least one pigeonhole has more than _____ pigeons.

Is there a hash function that avoids linear search complexity?

Theorem: Suppose a hash table of size m is used to store a set S of n keys drawn from the universe U , where $|U| > nm$. Then, *no matter which hash function* $h : U \rightarrow \{0, 1, 2, \dots, m-1\}$ is chosen, there is a set $S \subset U$ of n keys that all map to the same location.

Proof:

1. Pick any hash function of your choosing.
2. Map all keys of U using h to the table of size m .
3. By the pigeonhole principle, at least one table slot gets at least n keys because $|U| > nm$
4. Choose those n keys as input set S
5. Now h will map S to a single location.



Generalized Pigeonhole Principle:

If there are m pigeonholes and $nm + 1$ pigeons, at least one pigeonhole has more than n pigeons.

What does this mean for real-world applications?

Universal Hash Functions

Main result so far: No single hash function can avoid worst case linear time search complexity!

Main idea behind universal hash functions: Don't fix a single hash function. Choose one randomly from a “good” family of hash functions.

- Imagine an adversary picks **two keys**, x and y .
- You, the algorithm designer, get to **randomly choose** a hash function h from a big family H .
- The hash function is designed so that:
The chance that $h(x) = h(y)$ is **at most $1/m$** .

Example: $h(x) = ax + b \pmod{p}$ where p is a prime number

Previously: Balanced BST

```
set<string> groceries;  
groceries.insert("Banana") ;  
groceries.insert("Apple");  
groceries.insert("Milk");  
groceries.insert("Bread");
```

std::set

stores unique keys

```
map<string, int> groceries;  
groceries["Banana"] = 2;  
groceries["Apple"] = 1;  
groceries["Milk"] = 3;  
groceries["Bread"] = 5;
```

std::map

stores unique keys

Each key has an associated value

Today: Hash table and hash map

```
unordered_set<string> groceries;  
groceries.insert("Banana") ;  
groceries.insert("Apple");  
groceries.insert("Milk");  
groceries.insert("Bread");
```

std::unordered_set
stores unique keys

```
unordered_map<string, int> groceries;  
groceries["Banana"] = 2;  
groceries["Apple"] = 1;  
groceries["Milk"] = 3;  
groceries["Bread"] = 5;
```

std::unordered_map
stores unique keys
Each key has an associated value

Operations: find, insert, erase

All operations $O(1)$ on average

* no worst case guarantee but fast in practice

References

Professor Subhash Suri's CS 130A handout on hash tables:

<https://sites.cs.ucsb.edu/~suri/cs130a/Hashing.pdf>